SoK: The Quantifier Gap Between Certified Robustness and Airworthiness Assurance

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***Abstract*—Regulators now want quantitative safety evidence for machine learning: EASA expects it for learned components, the SORA risk assessment sets safety targets per flight hour, and ED-324 will codify learning assurance. Robustness research supplies guarantees, yet the two rarely meet, speaking about different things: a certified radius covers *one input*, a contingency volume *one flight*, and a per-flight-hour target a *population of flights*.**

We label every guarantee and target by what it covers and whether it is proved or statistical, and model the bridging arguments as *lifts*, steps that each owe evidence. A computed lift graph decides whether each guarantee meets each target directly, through assumptions needing their own evidence, or not at all. Four results follow. Proofs alone never reach a probability target. Per-input error budgets become hourly rates without assuming independence, but repeated confirmation needs it. SORA's test-hour ladder is exactly the zero-failure "rule of three". Runtime monitoring meets per-flight targets but moves the gap onto the monitor.

A registered review screens 4,000 of 25,373 records over three snowballing rounds and codes 587 papers against 22 objectives, estimating 6,300 eligible papers. The field is shifting to statistical guarantees, chiefly conformal prediction, and is not saturated: each round added papers but no new kind of guarantee. Only 21 % of lifted guarantees (14 %–24 % across coders) evidence the assumption their lift needs, and almost none reach per-flight-hour targets. A SORA evidence generator shows one sound end-to-end path. We certify no aircraft; we map where guarantees fall short.

***Index Terms*—certified robustness, airworthiness, learning assurance, runtime assurance, UAV, SORA, systematization of knowledge**

I. Introduction

An unmanned aerial vehicle (UAV) that navigates with a learned component now meets two kinds of guarantee. The first comes from the certified-robustness and formal-verification literature. It states that a neural network (NN) does not change its decision for any input within a radius of a given point [1]–[3]. It may also state that a closed loop with an NN controller stays inside a reachable tube [4], [5]. The second comes from aviation. It states that an operation keeps the aircraft inside its operational volume with a probability of failure below 10^{-3} or 10^{-4} per flight hour (FH) [6]. It also states that a learned constituent generalizes over its operational design domain (ODD) [7], [8]. The two communities increasingly cite each other. EASA's concept paper asks for robustness and generalization evidence on machine-learning (ML) constituents [7]. The joint standard ED-324/ARP6983 of the European Organisation for Civil Aviation Equipment (EUROCAE) and SAE International will turn that request into objectives [9]. The verification community, for its part, motivates its benchmarks with airborne collision avoidance (ACAS Xu) and aircraft taxiing [2], [10].

The observation: Citation is not transfer. The two kinds of statement rarely support one another, because they quantify over different objects. A certified radius is a statement about *every perturbation of one input*. A reachable tube is a statement about *every disturbance along one trajectory over a horizon*. A contingency volume is a statement about *one operation*. A safety objective per flight hour is a statement about *a population of operations*. The gap is not only in scope but also in modality. "For all" and "with probability at least $1-\delta$" are different claims. So are "with probability over the certifier's own random noise" and "with probability over the operations the aircraft will fly", and the literature routinely conflates them. Evidence transfers from a guarantee to an objective only when the guarantee's quantifier covers the objective's. Otherwise an argument must bridge the two, such as a dynamics bound, a concentration inequality or a runtime monitor. We call such an argument a *lift*. A lift has a price: an assumption that itself needs evidence.

This paper: We systematize the certified-guarantee literature for learned perception, estimation and control by what each guarantee quantifies over. We code aviation and UAS assurance objectives the same way, and we compute which transfers are sound, which need which lifts, and which are category errors. Our contributions are as follows.

- **A quantifier type system and a lift calculus** (Section III). Every guarantee and objective receives a type $\langle \text{scope}, \text{modality} \rangle$. Seven lifts are typed edges between types, and each lift emits named obligations with an evidence kind. Dominance is a partial order (Proposition 1), and transfer along a path is sound with a union-bound confidence (Theorem 1).
- **Four structural results.** No modality comes for free: every path from a "for all" guarantee to a probability objective crosses a statistical obligation (Proposition 2). The rate lift needs no independence assumption, but confirmation credit does (Proposition 3). The SORA 2.5 Annex E test-hour ladder is the zero-failure rule of three, and testing per frame buys no economy over testing per hour (Proposition 4). Runtime assurance relocates the gap to its monitor rather than closing it (Proposition 5).
- **A registered systematization protocol** (Section IV). It specifies block search strings, two inclusion tiers, a snow-

balling stop rule, an eight-field codebook, per-field reliability with Cohen's κ, Krippendorff's α, Gwet's AC1 and the prevalence-adjusted bias-adjusted kappa (PABAK), and a disclosed amendment log for every deviation, including screening and coding by large language model (LLM) instances.

- **A computed transfer table and a full study** (Sections VII and IX). Stratified probability samples of the 25,373 searched records and of three snowballing iterations give weighted estimates with intervals for the searched corpus and its citation neighbourhood. The table types 32 class states, 13 of them classes added to the pilot's 17, against six objective types: 14 discharges, 120 lifts and 58 category errors. No class discharges a per-flight-hour objective without a lift. One pilot finding is reversed: vehicle-facing perception guarantees are mostly per input. A second, that statistical guarantees for perception are not the least developed, holds in the searched corpus but not in the expanded field, where perception and control tie.
- **An evidence generator** (Section VIII). From the committed results of a published composition certificate [11], it emits a SORA Annex A-shaped evidence pack in which every lift's assumption appears as an obligation with its status.

Scope and honesty: Three boundaries belong with these claims. First, the transfer table is computed from a calculus that we define. Its verdicts are only as good as the lift rules, which we therefore state in full (Table II) and release as code. Second, every screening and coding decision was made by LLM instances working independently from a frozen codebook, with full texts limited to open-access copies (Section IV-F). The reliability we report therefore measures the clarity of the codebook, not agreement between experts. Every number is a macro generated from committed result files, so a human re-coding regenerates the paper. Third, nothing in this paper certifies an aircraft. We map mathematical guarantees onto an evidence framework. Several regulatory documents we code were in flux at our cut-off date (7 October 2026), and we key our claims to stable concepts rather than clause numbers.

II. Preliminaries and Problem Statement

A. Two Vocabularies

Certified guarantees: A *certified guarantee* about a learned component f is a sound statement, or a statistically valid one, of the form "for every x in a set S, property $\varphi(f,x)$ holds". It may also take the form "with probability at least $1-\delta$ over a stated distribution, φ holds". Robustness certificates set $S=\{x' : \|x'-x_0\| \le r\}$ and φ to "the decision does not change". They are computed by complete solvers based on satisfiability modulo theories (SMT) or mixed-integer programming [2], [12]. They are also computed by sound bound propagation, as in interval bound propagation (IBP) and CROWN [3], [13], [14], by abstract interpretation [15], [16], and by randomized smoothing (RS), whose certificate holds with confidence $1-\alpha$ over the certifier's own Monte Carlo noise [1]. Closed-loop verification of NN control systems (NNCS) sets S to an initial-state set and φ to "the trajectory stays in a safe set over horizon T" [4], [5], [17].

Assurance objectives: An *assurance objective* is a requirement that an applicant must show is met. SORA 2.5 of the Joint Authorities for Rulemaking of Unmanned Systems (JARUS) sets a target level of safety (TLS) of fewer than 10^{-6} fatalities per flight hour [18]. Its Annex E requires the probability of leaving the operational volume to be below 10^{-3}/FH for low and medium robustness and below 10^{-4}/FH for high robustness [6]. Its Annex A defines the operational volume as the flight geography plus a *contingency volume*, sized from navigation accuracy, position-holding error, reaction distance and the contingency manoeuvre [19]. Manned-aircraft system-safety guidance states catastrophic failure conditions per flight hour, down to 10^{-9} [20]. EASA's learning assurance asks for generalization bounds over the ODD and for robustness of the model [7]. The ASTM F3269 practice for run-time assurance (RTA) bounds a complex function's behaviour on every execution with a monitor and a recovery function [21].

B. The Running Example: A Composition Certificate

A recent composition [11] bounds the mission-completion rate (MCR) of a UAV against an adversary who delays relayed navigation messages just below an intrusion detector's threshold θ. Its chain is

$$\theta \mapsto \Delta(\theta) \mapsto \delta = \gamma\Delta \mapsto \rho = \delta\, e^{LT_c} \le m \mapsto \mathrm{MCR} \ge f(\delta), \tag{1}$$

where $\Delta(\theta) \le H\min(\theta, s)$ is the hidden delay over H compromised hops with contact-window slack s. Here γ is a measured delay-to-position rate, L a Lipschitz constant of the navigation dynamics, T_c a re-anchoring horizon of the extended Kalman filter (EKF), and m a corridor margin. A follow-on paper [22] treats Δ and γ as design variables traded through authentication. The parent's three stated conditions, on dynamics, re-anchoring and interface transfer, are exactly the assumptions of the lifts in Eq. (1). That is why it serves as our running example rather than our subject.

C. Problem Statement

Let $\mathscr{G}$ be the set of certified guarantees about learned components of autonomous vehicles published from 2017 to 2026, and let $\mathscr{J}$ be the set of assurance objectives stated by aviation and UAS regulation and standards at a fixed cut-off date. For a guarantee $g \in \mathscr{G}$ and an objective $o \in \mathscr{J}$ we ask whether g *discharges* o, whether g is *evidence for* o only through stated lifts, or whether citing g for o is a *category error*. We further ask which assumptions the lifts introduce, and whether the literature accounts for them. We test four hypotheses (H), each of which could fail.

H1 Most certified guarantees for learned *perception* quantify per input, and therefore discharge no assurance objective without a lift. We expect H1 to fail for learned *control*, where closed-loop reachability is mature.

H2 Lifts are used, but their assumptions are rarely accounted for as obligations that need evidence.

TABLE I
NOTATION.

$\tau = \langle \sigma, \mu \rangle$	quantifier type: scope σ, modality μ
$\mathsf{in}, \mathsf{tr}, \mathsf{op}$	one input; trajectories over a horizon; one operation
$\mathsf{pop_d}, \mathsf{pop_h}$	population per demand; rate per flight hour
$\forall, \forall_\alpha$	for all; for all with certifier confidence $1-\alpha$
$\mathbb{P}, \mathbb{E}, \exists$	probability over the environment; expectation; exists
ℓ, $\mathscr{O}(\ell)$	lift; its set of obligations
π, $\mathscr{O}(\pi)$	path of lifts; its obligations (assurance deficit)
δ_o, ε_o	failure probability of evidence for o; its error term
$\succeq$	dominance (obligation-free transfer)
f, p, λ	decision rate (Hz); per-decision hazard probability; rate per hour
k, ρ	confirmation length; error persistence $\Pr(e_t \mid e_{t-1})$
$\Delta, \gamma, L, T_c, m$	parent certificate: hidden delay, rate, Lipschitz constant, horizon, margin

H3 Per-flight-hour and ODD objectives are natural targets for statistical guarantees, yet statistical guarantees for learned perception are the least developed class.

H4 Runtime assurance is the dominant practical bridge for objectives that no offline certificate discharges.

Table I collects the notation. Fig. 1 shows how the answers are assembled.

III. Methodology I: A Quantifier Type System

The framework (Fig. 1) has five stages. This section develops stage (3), the theory against which the corpus is coded. Section IV develops stages (1), (2) and (5). Proofs omitted here are in Appendix A.

A. Types

Definition 1 (Quantifier type)**.** A guarantee or objective has type $\tau = \langle \sigma, \mu \rangle$. The *scope* $\sigma \in \{\mathsf{in}, \mathsf{tr}, \mathsf{op}, \mathsf{pop_d}, \mathsf{pop_h}\}$ names the object quantified over: inputs in a neighbourhood or region of one point (in), trajectories over a bounded horizon (tr), one complete operation or every execution (op), a distribution of demands such as inputs, environments or operations ($\mathsf{pop_d}$), or a rate per flight hour ($\mathsf{pop_h}$). The *modality* $\mu \in \{\forall, \forall_\alpha, \mathbb{P}, \mathbb{E}, \exists\}$ names the kind of claim: sound for all ($\forall$), for all with confidence $1-\alpha$ over the certifier's own randomness ($\forall_\alpha$), a probability bound over the environment or data ($\mathbb{P}$), a bound on an expectation ($\mathbb{E}$), or an existential witness ($\exists$).

Two distinctions in Definition 1 carry most of the weight. First, $\forall_\alpha$ is not $\mathbb{P}$. The confidence of randomized smoothing is over the certifier's coins. It says nothing about how often the aircraft meets an input outside the certified ball. Second, $\mathsf{pop_d}$ is not $\mathsf{pop_h}$. A misclassification probability per image, or a failure probability per mission, is a statement per demand. A target per flight hour also needs an exposure model. Fig. 2 draws the type plane.

B. Dominance

Definition 2 (Dominance)**.** Type τ *dominates* τ', written $\tau \succeq \tau'$, if a statement of type τ entails a statement of type τ' about the same property with no further evidence. A guarantee g *discharges* an objective o if $\tau_g \succeq \tau_o$ and the perturbation model of g covers the hazard of o.

Dominance is generated by two obligation-free rules. A probability bound on a failure event is a bound on the expectation of its indicator, so $\langle \sigma, \mathbb{P} \rangle \succeq \langle \sigma, \mathbb{E} \rangle$ for population scopes. A probability bound per operation is a statement about the population of operations, so $\langle \mathsf{op}, \mathbb{P} \rangle \succeq \langle \mathsf{pop_d}, \mathbb{P} \rangle$.

Proposition 1 (Dominance is a partial order)**.** *$\succeq$ is reflexive, transitive and antisymmetric. Under it, $\langle \mathsf{in}, \forall \rangle$ and $\langle \mathsf{pop_h}, \mathbb{P} \rangle$ are incomparable, and so are $\langle \mathsf{op}, \forall \rangle$ and $\langle \mathsf{pop_d}, \mathbb{P} \rangle$.*

The proposition is the formal content of the gap. Under $\succeq$, a radius is not a weaker or stronger kind of per-flight-hour claim. It is a different claim, and moving between the two is an argument, not a weakening. The second incomparability surprises at first: a per-operation "for all" looks stronger than any probability. It is stronger only on its certified set. To become a probability over deployed operations, it needs evidence that deployed operations stay in that set.

C. Lifts and Obligations

Definition 3 (Lift)**.** A *lift* $\ell : \tau \to \tau'$ is an inference rule whose premise is a statement of type τ and whose conclusion is a statement of type τ'. It is valid only together with a set of side conditions $\mathscr{O}(\ell)$, its *obligations*. Each obligation has an evidence kind: *proof*, *model*, *statistical*, or *architectural*.

Table II lists the seven lifts of the calculus. Each one corresponds to a family of arguments in the corpus. $\mathsf{L_{dyn}}$ is the Lipschitz–Grönwall step of the running example [23] and the reachability step of NNCS verification. $\mathsf{L_{op}}$ is the operational-profile argument of reliability assessment [24], [25]. $\mathsf{L_{sup}}$ is the argument, usually left implicit, that deployed operations stay inside the certified set. $\mathsf{L_{mon}}$ is the Simplex argument [26], [27].

D. Transfer Paths and Soundness

Definition 4 (Transfer path and assurance deficit)**.** A *transfer path* from guarantee g to objective o is a sequence of lifts $\pi = \ell_1 \cdots \ell_k$ with ℓ_1 starting at τ_g and ℓ_k ending at a type that dominates τ_o. Its *assurance deficit* is the obligation set $\mathscr{O}(\pi) = \bigcup_i \mathscr{O}(\ell_i)$, extended by $\mathsf{O_{thr}}$ when the hazard is not covered. The *verdict* is **D** (discharge) if some path has an empty deficit, $\mathbf{L}_k$ if the least deficit has k obligations, and $\times$ (category error) if no path exists.

Theorem 1 (Transfer soundness)**.** *Let g hold with confidence at least $1-\delta_0$, where $\delta_0 = 0$ for $\forall$ and $\delta_0 = \alpha$ for $\forall_\alpha$. Let π be a transfer path from g to o, and let each obligation $\mathsf{O} \in \mathscr{O}(\pi)$ be discharged by evidence that is valid with probability at least $1-\delta_{\mathsf{O}}$, where $\delta_{\mathsf{O}} = 0$ for proofs. Then the conclusion of π holds with confidence at least*

$$1-\delta_\pi, \qquad \delta_\pi \;=\; \delta_0 + \sum_{\mathsf{O} \in \mathscr{O}(\pi)} \delta_{\mathsf{O}}, \tag{2}$$

and its error term composes along the rules of Table II, additively through $\mathsf{L_{op}}$ and $\mathsf{L_{sup}}$ and multiplicatively by n_d through $\mathsf{L_{rate}}$. If the conclusion's type dominates τ_o, the objective's statement holds at that confidence.

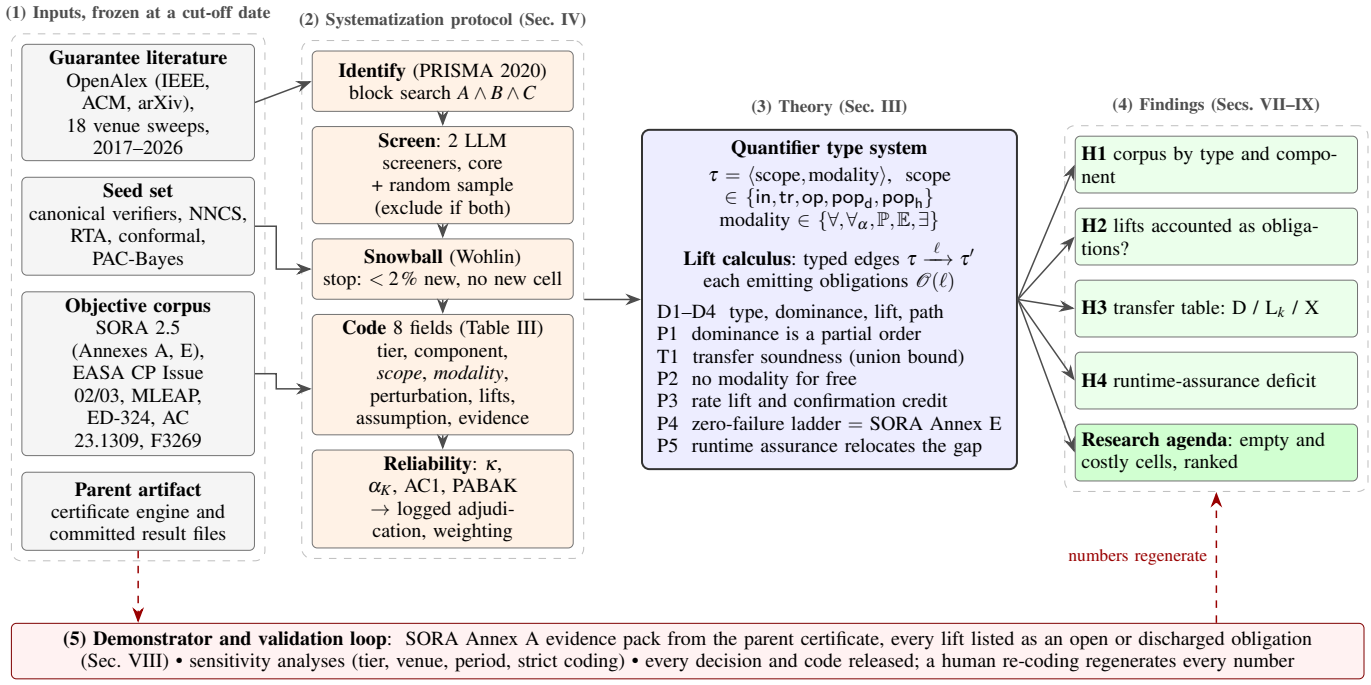


Fig. 1. Methodological framework. (1) Two corpora enter the study: guarantees about learned components, and assurance objectives from aviation regulation and standards, both frozen at a cut-off date. (2) A registered protocol identifies, screens, snowballs and codes the guarantees. Two LLM coders work independently from a frozen codebook, reliability is reported per field before adjudication, and corpus-level results are weighted estimates from a stratified sample. (3) Every guarantee and objective receives a quantifier type, and a calculus of typed lifts computes, for each pair, whether the guarantee discharges the objective, needs lifts whose assumptions become obligations, or is a category error. (4) The coded corpus and the calculus answer H1–H4 and yield a ranked research agenda. (5) An evidence generator instantiates one sound path end to end, and the validation loop regenerates every number from result files.

Theorem 1 is elementary, a union bound over the events on which some premise fails. Its use is accounting. A paper that lifts a guarantee without naming δ_O for each obligation has not established Eq. (2). It has established a conditional statement whose conditions an assurance case must carry. This is H2 made measurable: the coding scheme records, per guarantee, whether the assumption of each lift is unstated, stated, stated and evidenced, or evidenced with a confidence that enters the final claim.

E. Four Structural Consequences

Proposition 2 (No modality for free)**.** *Every transfer path from a type with modality $\forall$ or $\forall_\alpha$ to a type with modality $\mathbb{P}$ contains at least one lift with a statistical obligation.*

The proposition states something the field often argues informally [28], [29], and the calculus makes it exact. No chain of proofs, however sound or tight, turns a radius into a probability over deployment. At some point an argument must sample the world the aircraft will fly in. Better certificates shrink the other terms of Eq. (2). They cannot remove the statistical one.

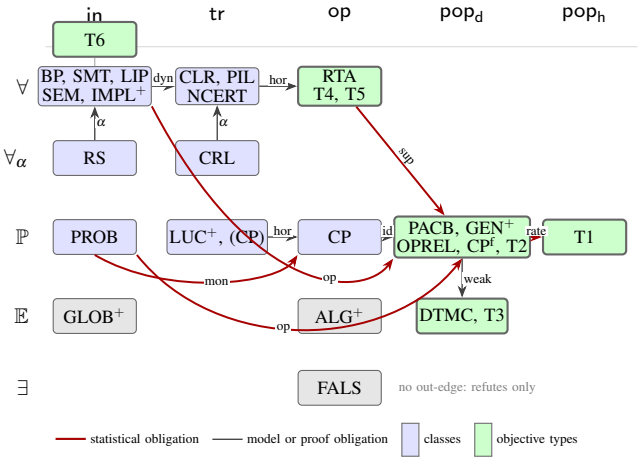


Fig. 2. The type plane. Columns are scopes, rows are modalities, and boxes place the guarantee classes ($^+$ added in the full study, $^\text{f}$ a field variant) and objective types T1–T6. Each arrow is a lift with obligations (Table II). Red arrows carry a statistical obligation, and every path from a $\forall$ state to a $\mathbb{P}$ state crosses one (Proposition 2). No arrow points left, and none leaves $\exists$ or $\mathbb{E}$.

Proposition 3 (Rate lift and confirmation credit)**.** *Let a component take f decisions per second, each hazardous with stationary marginal probability at most p. (a) The expected number of hazardous decisions per flight hour is at most $3600fp$, whatever the dependence between decisions. Hence a per-hour target λ is met whenever $p \le p^\star = \lambda/(3600f)$. (b) Suppose a hazard requires $k \ge 2$ consecutive erroneous decisions. Under arbitrary stationary dependence, the rate of hazardous runs is at most $3600fp/k$. Under a two-state Markov error process with persistence $\rho = \Pr(e_t \mid e_{t-1})$, it equals $3600fp(1-\rho)\rho^{k-1} \le 3600fp\,(k-1)^{k-1}/k^k$. Under*

TABLE II

THE LIFT CALCULUS. EACH LIFT IS AN INFERENCE RULE FROM A PREMISE TYPE TO A CONCLUSION TYPE, VALID TOGETHER WITH ITS OBLIGATIONS. KIND: P PROOF, M MODEL, S STATISTICAL, A ARCHITECTURAL. THE RULES ARE IMPLEMENTED AS THE EDGE SET OF `QUANTIFIER.PY`, AND THE PROPOSITIONS ARE CHECKED ON IT MECHANICALLY.

Lift	Edge	Rule (premise $\Rightarrow$ conclusion)	Obligations	Kind
L_{α}	$\langle \sigma, \forall_{\alpha} \rangle \to \langle \sigma, \forall \rangle$	certificate valid with probability $1-\alpha$ over the certifier's coins $\Rightarrow$ certificate valid, at confidence $1-\alpha$	O_{α}: accept α	P
$\mathsf{L}_{\mathrm{dyn}}$	$\langle \mathsf{in}, \forall \rangle \to \langle \mathsf{tr}, \forall \rangle$	$\|\delta u(t)\| \le r$ for all t, drift L-Lipschitz $\Rightarrow$ $\|e(t)\| \le r(e^{Lt}-1)$ on $[0, T_c]$	$\mathsf{O}_{\mathrm{lip}}$: L and model error over the operating region	M
$\mathsf{L}_{\mathrm{hor}}$	$\langle \mathsf{tr}, \forall \rangle \to \langle \mathsf{op}, \forall \rangle$; $\langle \mathsf{tr}, \mathbb{P} \rangle \to \langle \mathsf{op}, \mathbb{P} \rangle$	tube bound per window, re-anchoring every T_c $\Rightarrow$ bound on $\sup_{[0,T]} \|e\|$	$\mathsf{O}_{\mathrm{hor}}$: re-anchoring or horizon covers the operation	M
L_{op}	$\langle \mathsf{in}, \forall \mid \mathbb{P} \rangle \to \langle \mathsf{pop}_{\mathsf{d}}, \mathbb{P} \rangle$	robust on cells $c \in \mathscr{C} \Rightarrow \Pr_{x \sim \mathrm{OP}}[\mathrm{error}] \le \mathrm{OP}(\mathscr{X} \setminus \cup \mathscr{C}) + \varepsilon_{\mathrm{prof}}$	$\mathsf{O}_{\mathrm{prof}}$: operational profile and per-cell robustness	S
$\mathsf{L}_{\mathrm{sup}}$	$\langle \mathsf{op}, \forall \rangle \to \langle \mathsf{pop}_{\mathsf{d}}, \mathbb{P} \rangle$	$\varphi(\omega)$ for all $\omega \in S \Rightarrow \Pr_{\omega \sim D}[\neg\varphi] \le \Pr_D[\omega \notin S] \le \varepsilon_{\mathrm{odd}}$	$\mathsf{O}_{\mathrm{odd}}$: deployment stays in the certified set	S
$\mathsf{L}_{\mathrm{rate}}$	$\langle \mathsf{pop}_{\mathsf{d}}, \mathbb{P} \rangle \to \langle \mathsf{pop}_{\mathsf{h}}, \mathbb{P} \rangle$	per-demand hazard $\le p$, n_d demands per hour $\Rightarrow$ $\lambda \le n_d p$ (Prop. 3)	$\mathsf{O}_{\mathrm{dep}}$: exposure n_d, stationarity; dependence model for confirmation credit	S
$\mathsf{L}_{\mathrm{mon}}$	$\langle \mathsf{in}, \forall \mid \mathbb{P} \rangle \to \langle \mathsf{op}, \mathbb{P} \rangle$	monitor flags hazardous inputs, switch within latency, verified recovery $\Rightarrow$ per-operation safety except on monitor misses	$\mathsf{O}_{\mathrm{mcov}}$, $\mathsf{O}_{\mathrm{lat}}$, $\mathsf{O}_{\mathrm{rec}}$, $\mathsf{O}_{\mathrm{ind}}$	S,M,P,A

Free edges (no obligation): $\langle \mathsf{op}, \mathbb{P} \rangle \to \langle \mathsf{pop}_{\mathsf{d}}, \mathbb{P} \rangle$ (identity of reading); $\langle \mathsf{pop}_{\cdot}, \mathbb{P} \rangle \to \langle \mathsf{pop}_{\cdot}, \mathbb{E} \rangle$ (indicator expectation).
Hazard coverage: $\mathsf{O}_{\mathrm{thr}}$ is added whenever the guarantee's perturbation model does not cover the objective's hazard (natural, adversarial, or both).

independence it is approximately $3600 f p^k$.

At 30 Hz and 10^{-4}/FH, part (a) gives $p^\star = 9.3 \times 10^{-10}$ per frame (108,000 decisions per hour). At 10^{-9}/FH it gives $p^\star = 9.3 \times 10^{-15}$. Part (b) explains a common move. Temporal filtering is credited with turning a per-frame error of 9.7×10^{-4} into a per-hour rate of 10^{-4}, a gain of about 1×10^6. That credit is real only under independence. A first-order Markov process guarantees a gain of only $k^k/(k-1)^{k-1} = 6.75$ for $k = 3$, and arbitrary dependence guarantees only k (Fig. 4(b)). The obligation $\mathsf{O}_{\mathrm{dep}}$ is therefore where perception-based per-hour arguments stand or fall.

Proposition 4 (Zero-failure ladder)**.** *(a) Under a homogeneous Poisson failure process, T failure-free hours reject $\lambda \ge \lambda_0$ at level a if and only if $T \ge \ln(1/a)/\lambda_0$. At $a = 0.05$ the SORA 2.5 Annex E test durations of 30, 300, 3000 and 30 000 h [6] correspond to $\lambda_0 = 10^{-1}, \ldots, 10^{-4}$/FH within 0.15 %. (b) Demonstrating $p \le \lambda_0/(3600 f)$ per frame by zero-failure testing requires exactly as many hours of data as (a). It also replaces an independence assumption between hours with one between frames, which is harder to defend.*

Part (a) shows that a regulator already accepts a concentration lift, the "rule of three" [30], with explicit arithmetic. We use it as the worked example of a regulator-endorsed $\langle \mathsf{pop}_{\mathsf{d}}, \mathbb{P} \rangle \to \langle \mathsf{pop}_{\mathsf{h}}, \mathbb{P} \rangle$ step. Part (b) closes a tempting shortcut: 10^{-4}/FH at 30 Hz needs 3.2×10^9 failure-free frames, or 29,957 h, and 10^{-7}/FH needs about 3,417 years of continuous flight [31], [32].

Proposition 5 (Runtime assurance relocates the gap)**.** *In the calculus, $\langle \mathsf{op}, \forall \rangle$ objectives are discharged directly only by a guarantee of type $\langle \mathsf{op}, \forall \rangle$. Such a guarantee from runtime assurance is conditional on $\{\mathsf{O}_{\mathrm{mcov}}, \mathsf{O}_{\mathrm{lat}}, \mathsf{O}_{\mathrm{rec}}, \mathsf{O}_{\mathrm{ind}}\}$. When the monitor's decision depends on a learned perception output, $\mathsf{O}_{\mathrm{mcov}}$ is a statement of type $\langle \mathsf{pop}_{\mathsf{d}}, \mathbb{P} \rangle$ about that output, the same type as the gap the architecture was introduced to bridge.*

Corollary 1 (The cheapest route skips the operation)**.** *From $\langle \mathsf{in}, \forall \rangle$, the least-obligation route to $\langle \mathsf{pop}_{\mathsf{h}}, \mathbb{P} \rangle$ is $\mathsf{L}_{\mathrm{op}}\mathsf{L}_{\mathrm{rate}}$, with two obligations. It never visits $\langle \mathsf{op}, \forall \rangle$. Every route that also yields a per-operation guarantee costs at least four obligations: $\mathsf{O}_{\mathrm{lip}}$, $\mathsf{O}_{\mathrm{hor}}$, $\mathsf{O}_{\mathrm{odd}}$ and $\mathsf{O}_{\mathrm{dep}}$.*

The corollary separates two things an applicant may want. One is a per-hour number. The other is a per-operation envelope, which is what sizes a contingency volume. The cheaper route yields the first without the second. The running example takes the longer route and pays for it, and Section VIII shows what it buys.

IV. METHODOLOGY II: SYSTEMATIZATION PROTOCOL

A. The Guarantee Corpus

Sources and search: The protocol follows the systematic-review guidelines of Kitchenham and Charters [33] and reports with the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) 2020 [34]. We search OpenAlex [35] for 2017–2026, retrieving every result of every query. OpenAlex indexes the Crossref records of publications in IEEE Xplore and the ACM Digital Library, together with arXiv and DBLP, so one search reaches most of what those libraries hold; we did not run separate searches of the subscription databases Scopus and Web of Science. Each query conjoins three blocks: A, a guarantee term (16 phrases, from "certified robustness" and "reachability analysis" to "conformal prediction", "PAC-Bayes" and "runtime assurance"); B, a learning term; and C, an autonomy term (Appendix B). Through OpenAlex we also sweep the complete 2017–2026 tables of contents of IEEE S&P, USENIX Security, CCS, NDSS, CAV, TACAS, HSCC, ICCPS, EMSOFT, NFM, SAFE-

COMP, DSN, NeurIPS, ICML, ICLR, CoRL, CDC and DASC, keeping papers whose title contains a block-A phrase. Records are merged when they share a DOI, or a title with publication years at most two apart, so a preprint and its published version count once. Every request is logged with a timestamp, and we report the search's sensitivity against the pilot set (Section IV-F): the share of its papers the search retrieves. The registered strings found 56 % of the pilot set. One revision, with terms chosen from the misses of one half of the pilot set only, widened block C to control and cyber-physical phrasings. It retrieves 63 % of the pilot set (54/86), 72 % of its tier-A papers and 67 % of the held-out half (29/43). The search identified 33,664 records, of which 8,291 were duplicates, leaving 25,373 candidates. A sweep of the venues' DBLP tables of contents could not be run, because DBLP blocks automated access; we record it as not executed, and venue coverage rests on the OpenAlex sweeps.

Inclusion: We include a paper if it states a formal, statistical or architectural guarantee about a learned component, or about a system containing one, and if the object its guarantee quantifies over can be identified. We exclude empirical attack studies without a guarantee, guarantees about non-learned components, and theses that duplicate a journal version.

Two tiers: The inclusion rule in our proposal required evaluation on an autonomous vehicle. Applied literally, it would exclude the canonical verifiers that define the guarantee classes, which are evaluated on image benchmarks. We therefore code two tiers. *Tier A* contains guarantees evaluated on a vehicle task, including standard vehicle benchmarks such as ACAS Xu, TaxiNet, adaptive cruise control and quadrotors. *Tier B* contains method anchors evaluated only on generic benchmarks. Every statistic is reported for tier A and for both tiers together, so that a reader can see how much of a finding is carried by the anchors.

Snowballing: We snowball backward and forward from the included set, following Wohlin [36]. We stop when two consecutive iterations each add fewer than 2 % new inclusions *and* no iteration adds a new cell of the type plane. The second condition matters more: an SoK can miss papers, but it should not miss a kind of guarantee. On a sampled corpus we apply the rule to estimates. The seeds are the eligible coded papers, and new candidates are screened, sampled and coded as in the main study, for at most three iterations. The three iterations added an estimated 644, 1,089 and 535 eligible papers (16 %, 23 % and 9 % of the field before each), so the corpus is *not saturated* in papers. None added a cell of the type plane: the citation graph leads to more papers of the kinds already found, mostly generic learning theory. Tier A is 29 % of the expanded field against 40 % of the searched corpus.

B. Codebook

The unit of coding is a guarantee, the headline claim of a paper. Table III lists the eight fields. Each uses a closed vocabulary. Three decision rules prevent the confusions that Section III warns against. Randomized-smoothing confidence is coded $\forall_\alpha$, never $\mathbb{P}$. Falsification is coded $\exists$, because it can only refute. A certificate checked by sampling over its domain is coded with L_{conc}, because sampling has entered the claim. The pilot added three clarifying rules, which froze the codebook at version 1.1 before the full study (Appendix C).

TABLE III
CODEBOOK (VERSION 1.1). ONE ROW PER GUARANTEE.

Field	Values
tier	A (vehicle task) / B (method anchor)
component	perception / estimation / control / planning / end-to-end
scope	in / tr / op / pop
modality	$\forall$ / $\forall_\alpha$ / $\mathbb{P}$ / $\mathbb{E}$ / $\exists$
perturbation	norm / semantic / patch / disturbance / distribution / poisoning / none
lifts (list)	L_{dyn}, L_{hor}, L_{gen}, L_{op}, L_{rate}, L_{conc}, L_{sup}, L_{mon}, none
assumption	U unstated / S stated / SE stated and evidenced / SEC evidenced with a confidence that enters the claim
evidence	proof / measured / simulated / modelled

Two codebook lifts are not edges of the calculus. L_{gen} (a generative, abstraction or contract surrogate for the real sensor) and L_{conc} (a concentration, PAC, conformal or scenario argument from samples) change how a guarantee is *obtained*, not its type. They enter the transfer table as intrinsic obligations of the class, O_{cov} and O_{iid}, alongside the obligations of the calculus edges a class's members already use.

C. Coders and Reliability

Two coders code every paper independently. We report, per field, percent agreement, Cohen's κ [37], Krippendorff's nominal α [38], Gwet's AC1 [39] and PABAK, each with a percentile bootstrap interval over papers. We accept $\alpha \geq 0.80$, and $\alpha \geq 0.667$ for tentative conclusions only. We report AC1 and PABAK because the corpus is expected to be skewed toward in. Under skew, κ and α can be low despite high agreement, the prevalence paradox [40]. The set-valued lifts field is scored per lift, as a binary variable, and by mean Jaccard similarity. A third coder adjudicates every disagreement, logs a rationale, and may not change a field on which the two coders agree.

The registered protocol used two human coders, with an LLM as a measured third coder that could never exclude a paper. No human coding effort was available for a corpus of this size, so the study deviates. Each deviation was written down and committed before the data it governs existed. Screening, full-text coding and adjudication are done by independent LLM sub-agent instances on two different models. Each reads only the codebook, the inclusion rule, the class list and its own batch, and never another coder's output. A paper is excluded at screening only when both screeners exclude it. Published evaluations of LLM screening report moderate-to-substantial agreement that depends strongly on the clarity of the criteria [41], [42]. We therefore report LLM–LLM agreement as a measure of codebook clarity, never as expert agreement, and we report every headline quantity from each coder's codes alone as well as after adjudication.

D. The Objective Corpus

We code 22 objectives from nine documents: the SORA 2.5 main body and Annexes A and E [6], [18], [19]; EASA's AI concept paper Issue 02 and Proposed Issue 03 [7], [43]; the final report of EASA's Machine Learning Application Approval (MLEAP) project [8]; ED-324/ARP6983 at the level of concepts [9]; the Federal Aviation Administration (FAA) advisory circular AC 23.1309-1E and EASA's acceptable means of compliance AMC 25.1309 [20], [44]; ASTM F3269-21 [21]; and DO-326A [45]. Of these, 14 were checked against the primary text. The others, marked † in Table IV, rest on secondary sources and are flagged for checking. Objectives with no behavioural quantifier, such as the structural-coverage objectives of DO-178C [46], are recorded but excluded from the transfer table: they constrain the process, not the behaviour. The corpus is frozen at 7 October 2026. ED-324 and the Proposed Issue 03 were moving at that date, so we code them only at the level of stable concepts.

E. Computing the Transfer Table

The objectives fall into six types, by type and hazard (Table VI): per flight hour (T1), generalization over the ODD (T2), average error (T3), per operation (T4), security threat condition (T5) and local robustness (T6). Each guarantee class receives the modal type and modal perturbation of its adjudicated members. Its *intrinsic* obligations are those of the lifts that at least half of its members use. For each class and objective type, we compute the least-obligation path with Dijkstra's algorithm over the edge set of Table II and add $\mathsf{O}_{\mathrm{thr}}$ when the hazard is not covered. Norm-bounded, patch, semantic and poisoning models cover adversarial hazards; disturbance and distribution models cover natural ones. One class-level override is documented: a Simplex guarantee holds for arbitrary behaviour of the monitored component, but not for attacks on the monitor's own inputs, so it covers natural hazards only. Pilot classes keep the state fixed in the pilot. Guarantees that fit no pilot class form the classes added in the full study, which take their state from their weighted full-study members by the same rule. When a pilot class's weighted modal type in the full study differs from its pilot type and it has at least five coded members, the table also shows a field variant of the class.

F. Pilot and Full Study

Pilot. A purposive seed set of 86 guarantees (46 in tier A, 40 in tier B) in 17 classes, drawn from the canonical systems and from a scoping review, fixed the codebook at version 1.1 and the class list. It is not a sample. We use it to validate the search and to define the pilot classes, and we report its findings only beside the full study.

Screening. A ranker trained on one half of the pilot set orders the 25,373 candidates (on the held-out half, 25 of 27 screened papers are included). Two LLM screeners judge title and abstract against the inclusion rule. The first 2,750 ranks form a core stratum screened in full, with 1,871 includes. Because the include rate had levelled off near 55 %, exhaustive screening was replaced, before any further data, by a simple random sample of 1,250 of the remaining 22,623 records. It includes 233 (19 %, 95 % interval $[17\,\%, 21\,\%]$), so the remainder holds about 4,217 includes ($[3{,}737, 4{,}731]$). The two screeners agree with $\kappa = 0.91$.

Full text and coding. A stratified random sample of the includes, allocated in proportion to each stratum's estimated includes, goes to full text. Only open-access copies are fetched. A record without one is replaced by the next in its stratum's random order: 25 of 117 core records and 38 of 233 remainder records were not retrieved. The 287 retrieved papers (92 core, 195 remainder) are coded for full-text eligibility and the eight fields, and 191 are eligible. Each coded paper stands for its stratum's estimated includes divided by its coded papers. Corpus-level results are weighted shares with 95 % bootstrap intervals that resample papers within strata and redraw the remainder's include count. They are estimates for the field, never counts.

Snowballing and the expanded field. Iterations follow references and citations through OpenAlex; candidates are screened in full up to 3,000 (else a ranked core plus a random sample), and up to 100 includes per iteration are coded as above. The expanded field combines all 6 strata (587 coded papers). We report it throughout, with the searched-corpus value where the two differ in substance.

G. The Evidence Generator

The generator reads a certified configuration of the running example from its committed result files. It emits a SORA Annex A-shaped pack in which: the ODD is the parent's measured operating region; the position-holding term of the contingency volume is the certified tube radius under the stated adversary; every other term of the contingency volume is an open obligation of the operator; and every lift of Eq. (1) appears, typed, with its obligation and status. The last point is the generator's contribution. A compliance pack that lists the certificate but hides its assumptions would commit exactly the category error this paper names.

V. The Guarantee Landscape

We now systematize what each level offers, with classes in parentheses; a $^{+}$ marks the classes the full study added and a * those added from the snowballing iterations. Appendix D lists pilot membership. By weighted estimate, 30 % of the expanded field's guarantees are per input, 17 % per trajectory, 10 % per operation and 44 % per population (39 % in the searched corpus). The levels are uneven in a second way. In the searched corpus the highest-ranked records, which resemble the pilot, are 69 % proofs, while the rest of the field is 13 % proofs and 44 % simulation.

A. Per Input

Per-input certificates are the most developed class. Complete methods decide properties over input regions exactly [2], [12], [47] (SMT). Incomplete sound methods scale through linear

relaxations, branch and bound, and cutting planes [3], [13]–[16], [48], [49] (BP). Lipschitz-constrained architectures make a certified margin a by-product of training [50]–[52] (LIP). Randomized smoothing scales certification to large models, including detection and segmentation [1], [53]–[57] (RS). Semantic, geometric and patch certificates replace the norm ball with a parametrized set of rotations, illumination changes or bounded patches [58]–[64] (SEM). These parametrized sets are the closest the literature comes to EASA's language of an ODD. Each set is still a neighbourhood of one input, not a distribution over the operations the aircraft will fly. The international verification-of-NN competition (VNN-COMP) standardizes this level: its specifications are pre- and postconditions on one network [65].

Aviation case studies sit here too. ACAS Xu properties are checked over regions of encounter geometry [2], [66]. Industrial studies have assessed an exact ACAS Xu implementation against draft ED-324 objectives, with formal verification discharging some learning-assurance objectives [67], [68]. The region quantifier makes these statements per region of state space, which is still per input in our sense. Lifting them to an encounter needs a closed-loop model. Two small classes sit beside these certificates. Global robustness obtained by Lipschitz-based optimization over the whole input domain is reported as an expected quantity over that domain, so its modal type is $\langle \mathsf{in}, \mathbb{E} \rangle$ rather than a $\forall$ bound (GLOB^{+}). Verification of the executable implementation, such as its timing, makes a $\forall$-claim about code rather than about the learned function (IMPL^{+}).

B. Per Trajectory

For learned *control* with low-dimensional state inputs, closed-loop verification is mature (CLR). Taylor-model, star-set, polynomial-zonotope and semidefinite methods compute reachable tubes for NN controllers in feedback with known plants [4], [5], [17], [69]–[78]. The ARCH-COMP category for AI and NN control systems (AINNCS) benchmarks them yearly [79]. Its 2025 report notes that tools abstracting the problem differently produced disagreeing results on two benchmarks, direct evidence that the quantified object itself can differ between tools. Neural Lyapunov and barrier certificates give $\forall$-trajectory statements over a domain [80]–[82] (NCERT). Certified reinforcement learning bounds actions or cumulative reward under bounded observation perturbations [83]–[86] (CRL). A growing line of work learns the model uncertainty itself, as Gaussian-process error bounds, chance constraints or adaptively estimated parameters, and certifies a controller against it. Its claims hold per trajectory with a probability (LUC^{+}).

For learned *perception* in the loop, every result rests on a surrogate (PIL). The surrogate is a generative model that maps states to plausible images [10], a learned abstraction [87], a perception contract [88]–[90], a camera model [91], or a composition of these [92]–[94]. The guarantee then covers the surrogate's range, not real images. A probabilistic analysis of Boeing's taxiing system states plainly that it is best viewed as average-case rather than worst-case [95] (DTMC). This is the pattern H2 predicts: a type-changing lift (L_{gen}) whose assumption, surrogate coverage, is of type $\langle \mathsf{pop}_{\mathsf{d}}, \mathbb{P} \rangle$.

C. Per Operation

Runtime assurance gives per-execution guarantees by switching from an unverified controller to a verified fallback when a monitor fires [26], [27], [96]–[99] (RTA). Shields enforce a safety specification on a learning agent's actions [100]. ASTM F3269 standardizes the architecture for aircraft and makes "RTA system coverage" an explicit concern [21]. Monitors for learned perception detect activation patterns or inputs outside the training distribution [101]–[103] (MON). Their detection rates are population statements, usually reported without confidence. Falsification searches mission specifications in signal temporal logic (STL) for counterexamples [104]–[107] (FALS). It is the per-operation counterpart of testing: it can refute, never discharge.

D. Per Population

Statistical guarantees bound generalization by PAC-Bayes [108]–[110], including for policies across novel environments [111] (PACB). Conformal prediction calibrates trajectory predictors and STL runtime verification with distribution-free coverage under exchangeability [112]–[116], and under bounded distribution shift [117], [118] (CP). Probabilistic robustness estimates the mass of a neighbourhood that is misclassified [119]–[121] (PROB). It is a population statement over a ball, not over deployment. Operational-profile reliability assessment is the one line of work that lifts per-input robustness to a probability of misclassification per demand. It then argues from that to system-level safety targets [24], [25], [122], [123], alongside rare-event simulation of end-to-end systems [124], [125] (OPREL). It is the literature's most explicit $\langle \mathsf{in}, \forall \rangle \to \langle \mathsf{pop}_{\mathsf{d}}, \mathbb{P} \rangle$ lift, and it states its assumptions. Even so, it treats them as modelling choices rather than as obligations with a confidence. The negative results on demonstrating ultra-high reliability by testing bound what any of these can achieve [28], [31], [32].

The full study puts the weight of the field at this level. Conformal prediction is the largest class, at 19 % of guarantees ($[15\,\%, 23\,\%]$; 26 % in the searched corpus). The pilot coded its members as per-operation runtime checks, but in the field the dominant form is marginal coverage over a population. Learning-theoretic generalization and sample-complexity bounds outside PAC-Bayes form the second largest class (GEN^{+}, 14 % against 7 % in the searched corpus), and convergence and regret bounds for the learning algorithm (ALG^{+}, 8 %) are guarantees about training, not about the deployed component. Snowballing grows both, and adds population-level robustness limits (RLIM*) that bound what any classifier can achieve rather than certify one.

VI. The Objective Landscape

Table IV codes the objective corpus. Three observations stand out.

Regulators quantify per flight hour, and they mix quantifiers inside one document. SORA's target levels of safety and containment criteria are $\langle \mathsf{pop_h}, \mathbb{P} \rangle$. Its operational and contingency volumes are $\langle \mathsf{op}, \forall \rangle$ geometry, and it states that the specific assurance and integrity level (SAIL) is not quantitative [18]. EASA asks for generalization bounds over the ODD ($\langle \mathsf{pop_d}, \mathbb{P} \rangle$) and robustness of the model ($\langle \mathsf{in}, \forall \rangle$) in one learning-assurance process. In the same document, it notes that generalization-based error estimates are quantities on average ($\langle \mathsf{pop_d}, \mathbb{E} \rangle$) [7].

The ODD defines the domain of the population quantifier. EASA distinguishes an operational domain (OD) at system level from an ODD at constituent level, the latter a refinement of the former [7]. Under our types, the ODD is the support of D in $\mathsf{L_{sup}}$ and the domain of OP in $\mathsf{L_{op}}$. Data completeness and representativeness objectives are therefore the evidence that discharges $\mathsf{O_{odd}}$ and $\mathsf{O_{prof}}$ [8].

Architectural mitigation is the regulators' own bridge. SORA's "no single failure" criterion, AC 23.1309's equivalent, ASTM F3269, and EASA's monitoring of inputs outside the ODD are all $\langle \mathsf{op}, \forall \rangle$ objectives that runtime assurance targets directly [6], [7], [20], [21]. Secondary commentary on the Proposed Issue 03 reports an independent operational oversight system for higher automation levels [43]. We record that as $\langle \mathsf{op}, \forall \rangle$ and flag it for checking against the primary text.

VII. Results

A. Reliability of the Coding

Table V reports agreement between the two coders on the searched sample. The fields the findings rest on are reliable: Krippendorff's α is 0.89 for the type ($[0.84, 0.94]$; pilot 0.83), 0.97 for the tier, 0.75 for eligibility and 0.74 for the open-vocabulary class. The assumption field shows the prevalence paradox (77 % agreement, $\kappa = 0.47$, AC1 0.73) and is the one field below $\alpha = 0.667$, so H2 is reported from each coder separately. An adjudicator resolved every disagreement without changing an agreed field. It ran on the second coder's model and took that coder's value in 166 of 219 field disagreements, a second reason to report each coder's estimate.

B. H1: Where Guarantees Sit

Fig. 3 shows the estimated expanded field by component and scope. Across both tiers, 53 % of perception guarantees quantify per input ($[45\,\%, 61\,\%]$, $n = 178$; 51 % in the searched corpus), against 18 % of control guarantees ($[11\,\%, 25\,\%]$). For control, 40 % quantify per trajectory. H1 fails for control, as predicted. For perception it holds only in a weak form: per input is the largest level, but it is about half, not most. The rest of perception is mostly population-level statistics. Each coder's codes alone give the same picture (53 % and 53 %).

The full study reverses the pilot's refinement of H1. In the pilot, only 14 % of tier-A perception guarantees, those evaluated on vehicles, were per input, and we read this as a shift of vehicle-facing perception to closed-loop verification through surrogates. In the expanded field, 58 % of tier-A perception guarantees are per input ($[38\,\%, 80\,\%]$; 58 % and 58 % by coder; 59 % in the searched corpus). Perception-in-the-loop verification is a small class, fewer than one guarantee in a hundred (about 22 papers). The pilot had over-sampled it by design. Vehicle-facing perception is therefore mostly certified per input, like perception in general, and the surrogate route to a trajectory claim is the exception.

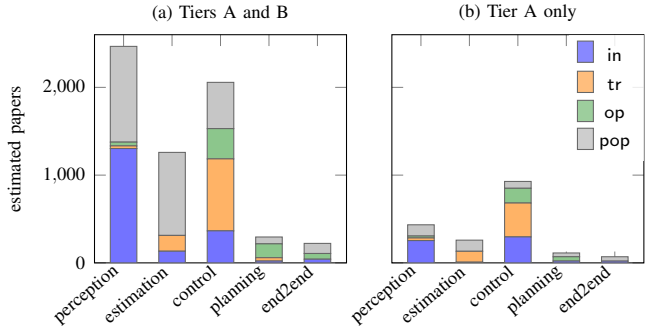


Fig. 3. The estimated field by component and scope: weighted estimates of the number of eligible papers in the expanded field (searched corpus and three snowballing iterations). (a) Across both tiers, perception guarantees are mostly per input or per population, and control guarantees mostly per trajectory. (b) Among guarantees evaluated on vehicles (tier A), perception is still mostly per input; closed-loop perception through surrogates, which dominated the pilot, is rare. Provenance: FULL STUDY, LLM CODING.

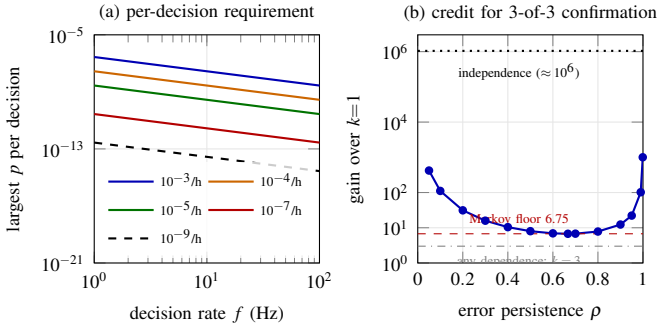


Fig. 4. The arithmetic of the per-flight-hour quantifier. (a) Largest per-decision hazard probability compatible with a per-hour target, $p^\star = \lambda/(3600 f)$ (Proposition 3(a)). At 30 Hz and 10^{-4}/h it is 9.3×10^{-10}. (b) What temporal confirmation buys at 30 Hz and 10^{-4}/h. Under independence the gain is about 10^6. Under a first-order Markov error process the gain is never below 6.75, whatever the persistence ρ, and under arbitrary stationary dependence it is only guaranteed to be $k = 3$ (Proposition 3(b)). Provenance: DERIVED (closed form).

C. H2: Are Lifts Accounted For?

An estimated 66 % of guarantees use at least one lift. Of these, 5 % leave the lift's assumption unstated and 74 % state it without evidence. 19 % evidence it, and only 2 % carry that evidence's confidence into the final claim. In total 21 % ($[16\,\%, 27\,\%]$; 28 % in the searched corpus) account for the assumption as an obligation in the sense of Theorem 1. This share rests on the least reliable field, and it depends on the coder: 14 % from the first coder's codes and 24 % from the second's. Every reading leaves a large majority of lifted guarantees unaccounted for, so H2 is supported. The pilot's 17 % lies inside that range.

TABLE IV

THE OBJECTIVE CORPUS, CODED BY TYPE AND HAZARD (N NATURAL, ADV ADVERSARIAL, B BOTH). †WORDING RESTS ON A SECONDARY SOURCE AND IS FLAGGED FOR CHECKING. GENERATED FROM `CORPUS/OBJECTIVES.CSV`.

ID	Document	Statement (paraphrased)	Type	Hazard	Target/FH
O01	JARUS SORA 2.5 Main Body (JAR-doc-25)	fewer than 1E-6 fatalities per flight hour	$\langle \mathsf{pop}_\mathsf{h}, \mathbb{P} \rangle$	N	1×10^{-6}
O02	JARUS SORA 2.5 Main Body (JAR-doc-25)	fewer than 1E-7 mid-air collisions per flight hour	$\langle \mathsf{pop}_\mathsf{h}, \mathbb{P} \rangle$	N	1×10^{-7}
O03	JARUS SORA 2.5 Annex E (JAR-doc-28)	probability of leaving the operational volume below 1E-3 per flight hour	$\langle \mathsf{pop}_\mathsf{h}, \mathbb{P} \rangle$	N	1×10^{-3}
O04	JARUS SORA 2.5 Annex E (JAR-doc-28)	probability of leaving the operational volume below 1E-4 per flight hour	$\langle \mathsf{pop}_\mathsf{h}, \mathbb{P} \rangle$	N	1×10^{-4}
O05	JARUS SORA 2.5 Annex E (JAR-doc-28)	no single failure leads to operation outside the ground risk buffer	$\langle \mathsf{op}, \forall \rangle$	N	–
O06	JARUS SORA 2.5 Main Body and Annex A (JAR-doc-26)†	contingency volume sized from GNSS accuracy and position-holding error and reaction and stopping distance; entry is an abnormal situation	$\langle \mathsf{op}, \forall \rangle$	N	–
O07	JARUS SORA 2.5 Annex E (JAR-doc-28)	30/300/3000/30000 flight hours for SAIL I-IV at 95 percent confidence	$\langle \mathsf{pop}_\mathsf{h}, \mathbb{P} \rangle$	N	–
O08	FAA AC 23.1309-1E	catastrophic failure condition below 1E-6 (Class I) to 1E-9 (Class IV) per flight hour	$\langle \mathsf{pop}_\mathsf{h}, \mathbb{P} \rangle$	N	1×10^{-9}
O09	FAA AC 23.1309-1E	no single failure results in a catastrophic failure condition	$\langle \mathsf{op}, \forall \rangle$	N	–
O10	EASA CS-25 AMC 25.1309†	catastrophic failure conditions of the order of 1E-9 per flight hour or less	$\langle \mathsf{pop}_\mathsf{h}, \mathbb{P} \rangle$	N	1×10^{-9}
O11	EASA AI Concept Paper Issue 02†	quantifiable generalisation bounds over the AI/ML constituent ODD	$\langle \mathsf{pop}_\mathsf{d}, \mathbb{P} \rangle$	N	–
O12	EASA AI Concept Paper Issue 02	generalisation-based error estimates are quantities on average	$\langle \mathsf{pop}_\mathsf{d}, \mathbb{E} \rangle$	N	–
O13	EASA AI Concept Paper Issue 02	data completeness and representativeness over the ODD as a prerequisite for generalisation bounds	$\langle \mathsf{pop}_\mathsf{d}, \mathbb{P} \rangle$	N	–
O14	EASA AI Concept Paper Issue 02†	stability of the model under perturbations and adverse conditions	$\langle \mathsf{in}, \forall \rangle$	B	–
O15	EASA AI Concept Paper Issue 02	monitoring to detect input data outside the AI/ML constituent ODD	$\langle \mathsf{op}, \forall \rangle$	N	–
O16	EASA AI Concept Paper Proposed Issue 03†	independent oversight system bounding Level 3 AI behaviour	$\langle \mathsf{op}, \forall \rangle$	N	–
O17	EASA MLEAP final report (2024)	generalisation relative to the ODD	$\langle \mathsf{pop}_\mathsf{d}, \mathbb{P} \rangle$	N	–
O18	EASA MLEAP final report (2024)	local robustness and stability of the trained model	$\langle \mathsf{in}, \forall \rangle$	B	–
O19	EUROCAE ED-324 / SAE ARP6983 (concept level)†	generalisation of the ML model	$\langle \mathsf{pop}_\mathsf{d}, \mathbb{P} \rangle$	N	–
O20	EUROCAE ED-324 / SAE ARP6983 (concept level)†	stability and robustness of the ML model	$\langle \mathsf{in}, \forall \rangle$	B	–
O21	ASTM F3269-21	safety monitor and recovery function bound the behaviour of a complex function on every execution	$\langle \mathsf{op}, \forall \rangle$	N	–
O22	RTCA DO-326A / EUROCAE ED-202A†	intentional unauthorised electronic interaction does not lead to an unacceptable threat condition	$\langle \mathsf{op}, \forall \rangle$	Adv	–

TABLE V

FULL-STUDY INTER-CODER RELIABILITY: PAPERS COMPARED, PERCENT AGREEMENT, COHEN'S κ, KRIPPENDORFF'S α WITH 95 % BOOTSTRAP INTERVAL, GWET'S AC1 AND PABAK. ELIGIBILITY IS OVER ALL CODED PAPERS; THE OTHER FIELDS OVER PAPERS BOTH CODERS FOUND ELIGIBLE. BOTH CODERS ARE INDEPENDENT LLM INSTANCES, SO THE TABLE MEASURES CODEBOOK CLARITY. GENERATED FROM `RESULTS/RELIABILITY_FULL_C1_FULL_C2.CSV`.

Field	n	%	κ	α [95 % CI]	AC1	PABAK
full-text eligibility	287	90	0.75	0.75 [0.66, 0.83]	0.82	0.79
class	186	76	0.74	0.74 [0.67, 0.80]	0.76	0.76
tier	186	98	0.97	0.97 [0.92, 1.00]	0.97	0.97
component	186	94	0.91	0.91 [0.86, 0.95]	0.92	0.92
scope	186	94	0.92	0.92 [0.87, 0.96]	0.92	0.92
modality	186	97	0.95	0.95 [0.91, 0.99]	0.97	0.96
type (scope×modality)	186	91	0.89	0.89 [0.84, 0.94]	0.91	0.91
perturbation	186	91	0.87	0.87 [0.81, 0.92]	0.89	0.89
assumption status	186	77	0.47	0.47 [0.31, 0.59]	0.73	0.69
evidence class	186	79	0.70	0.70 [0.62, 0.78]	0.73	0.72

The rate differs by lift. The concentration lift is the most common (47 % of guarantees, as conformal and PAC arguments dominate) and is evidenced in 22 % of its uses; the dynamics lift (12 %) in only 13 %, the plant model taken as given; the monitor lift (9 %) in 35 %. The support lift is nearly absent (3 %), although every per-operation guarantee (10 % of the field) needs it: the lift most often left implicit is the one regulators care most about.

The pilot suggested a trend, from 7 % accounted in 2017–2021 to 26 % in 2022–2026. The searched corpus does not confirm it (23 % against 29 %, intervals overlapping). The expanded field shows it (11 %, [5 %, 18 %], against 27 %, [20 %, 35 %]), as do the second coder's codes alone, but not the first coder's (10 % against 16 %, intervals overlapping). Since a conclusion needs all readings, the trend is suggestive, not established.

D. H3: The Transfer Table

Table VI is the computed transfer table on the expanded field: the 17 pilot classes, the 5 classes the full study added, the 8 classes grouped from the snowballing iterations, and 2 field variants. Of its 192 cells, 14 are discharges, 120 need lifts, and 58 are category errors. Four patterns stand out.

No class discharges a per-flight-hour objective (T1). 27 of the 32 rows reach T1, all through L_{rate}, so no per-hour claim exists without exposure and dependence evidence. The cheapest classes (CP, PACB, OPREL and the added GEN and ALG) need only that one lift, which matches H3's prediction that statistical guarantees are the natural fit. Every per-input certificate class needs three or four obligations. Proposition 2 is visible in the column: every T1 path carries a statistical obligation. The field variants change little. Conformal prediction's population form has the same row as its per-operation form, and certified reinforcement learning with $\forall$ rather than $\forall_\alpha$ claims saves the α lift.

Statistical guarantees are common; the rate lift is not. The second half of H3, that statistical guarantees for perception are the least developed, is reversed in the searched corpus (38 % of perception guarantees have modality $\mathbb{P}$, against 24 % for control, in both coders' codes). In the expanded field perception and control tie (36 %, [28 %, 43 %], and 36 %) and

the coders split, so we neither confirm nor reverse it. Both estimates agree that estimation, mostly conformal trajectory prediction, leads at 80 %, and that the scarce step is the one that turns statistical guarantees into airworthiness evidence. Only 1 % of guarantees ($[0\,\%, 2\,\%]$) perform the rate lift to a per-hour or per-demand system figure. The first half of H3, that per-hour and ODD objectives are the natural targets of statistical guarantees, is supported by the table.

Local robustness objectives (T6) are never discharged. EASA and MLEAP robustness refers to both natural and adversarial perturbations. Every per-input class covers one of them, so every cell carries O_{thr}. This is a modelling gap, not a proof gap: the certified sets of the literature were designed for an adversary that the objective does not single out.

Falsification and average-case results discharge nothing above their own type. 3 rows are all $\times$: falsification, global robustness estimates and differential privacy of training (DPG*). DTMC discharges only the average-error objective T3, and the population robustness limits (RLIM*) reach only T3. This is the formal version of a familiar warning, that a failed search for counterexamples or an average case is not evidence for a worst-case or tail objective. No coded full-study paper falls into the falsification class.

The calculus types claims, not their subject. The convergence and regret bounds of ALG$^+$ were all $\times$ in the searched corpus. In the expanded field their modal type is $\langle \mathsf{pop}_\mathsf{d}, \mathbb{P} \rangle$ (50 coded papers), so the table lets them discharge T2 and T3. That is right about the type and wrong about the subject, the training algorithm rather than the deployed model. We report the row as computed; it marks a missing lift, from an algorithm to the model it returns.

E. H4: Runtime Assurance

Of the 32 rows, 19 reach a per-operation objective (T4), and only RTA discharges one directly. This is H4's prediction, but Proposition 5 qualifies it. RTA carries 4 intrinsic obligations: monitor coverage, switching latency, verified recovery and independence. RTA is 8 % of the expanded field ($[5\,\%, 11\,\%]$; 11 % of the searched corpus), and 96 % of its guarantees rest on the monitor lift. In the pilot no RTA paper evidenced its assumption. In the full study 31 % do, but the interval is wide ($[12\,\%, 52\,\%]$) and the coders differ (11 % and 29 %); the codebook does not record the four obligations separately. For a perception-based monitor, coverage is a $\langle \mathsf{pop}_\mathsf{d}, \mathbb{P} \rangle$ statement about the monitor's learned input, so the architecture has moved the original gap onto its own monitor. H4 is supported in the form "RTA is the only direct bridge, and it relocates the deficit". It is not supported in the stronger form "RTA closes the gap". The independence obligation is where the two forms part. If the monitor reads the same perception channel it guards, the failures are common-cause and the lift collapses.

F. What the Per-Flight-Hour Quantifier Costs

Fig. 4 turns Propositions 3 and 4 into numbers. Panel (a) is the budget that a per-frame argument must meet. At 30 Hz it is 9.3×10^{-10} for 10^{-4}/FH, and 9.3×10^{-15} for 10^{-9}/FH. No certified-radius result in the corpus speaks to a quantity of this kind, because a radius bounds no probability. Panel (b) shows why the usual escape, temporal confirmation, needs O_{dep}. Without independence evidence, three-frame confirmation is guaranteed to buy between 3 and $6.75\times$, not the $10^6\times$ that the independence calculation suggests.

VIII. Case Study: An Evidence Pack for the Running Example

We apply the generator to the composition certificate at its operating point: detector setting $\theta = 0.25$ s, $H = 2$ hops, re-anchoring horizon $T_c = 1$ s and measured local Lipschitz constant $L = 1.181$. Table VII shows the typed chain and the pack's obligations.

What the pack certifies: With the empirically calibrated rate $\gamma = 1.365$ m/s, the induced perturbation is 0.68 m, and the tube radius is $\rho = 2.22$ m. Taking the supremum rate over all calibration samples ($\gamma = 1.625$ m/s) gives 2.65 m. The position-holding term of the SORA contingency volume under the stated adversary is therefore certified at the 5, 10 and 20 m margins, and not at 2 m. Under the kinematic worst case it would be 24.4 m, and nothing would be certified. The pack writes the other contingency-volume terms (navigation accuracy, map error, reaction distance, contingency manoeuvre) as operator obligations, because the certificate says nothing about them.

What it does not certify: Of the pack's 6 obligations, 2 are open or assumed. The chain is a valid transfer path to $\langle \mathsf{op}, \forall \rangle$ (T4) and, through L_{sup}, to $\langle \mathsf{pop}_\mathsf{d}, \mathbb{P} \rangle$ (T2). It stops there: O_{dep} is open, so the pack supports no per-flight-hour claim, and it says so. As Corollary 1 predicts, the parent pays four obligations where two would buy a per-hour number, and obtains the per-operation envelope that sizes a contingency volume.

Why this reframes a limitation: The parent bounds a proportion of missions, not a single flight. In the type plane that is the correct level for an airworthiness argument: every $\langle \mathsf{pop}_\mathsf{h}, \mathbb{P} \rangle$ objective is a population statement. This is a reframing, and it does not close the parent's interface-calibration limitation, the pack's only partially evidenced obligation.

IX. Research Agenda

The empty and costly cells of Table VI define the agenda. We rank them by how many objectives they would discharge.

1) **Evidence for** O_{dep} (all T1 cells, 7 objectives). Per-hour objectives are reachable from every non-refuting class only through the rate lift, and only 1 % of the field performs it. Temporal-dependence models of perception error, with confidence and validated on flight data, are the single most valuable missing piece. Proposition 3 shows that credit for confirmation turns on them. The field's statistical guarantees, conformal coverage above all, are one lift away from T1 and wait on exactly this evidence.
2) **Reachable tubes as contingency volumes** (T4, 6 objectives). Q_{tr} tools could size SORA's position-holding term

TABLE VI

THE COMPUTED TRANSFER TABLE ON THE EXPANDED FIELD. ROWS ARE GUARANTEE CLASSES WITH THEIR ESTIMATED NUMBER OF PAPERS AND THEIR TYPE: PILOT CLASSES WITH THE TYPE FIXED IN THE PILOT, ^{f} A PILOT CLASS'S FULL-STUDY MODAL TYPE WHERE IT DIFFERS, $^{+}$ CLASSES THE FULL STUDY ADDED AND * CLASSES GROUPED FROM THE SNOWBALLING ITERATIONS. COLUMNS ARE OBJECTIVE TYPES: T1 PER FLIGHT HOUR $\langle \mathsf{pop_h}, \mathbb{P} \rangle$, T2 ODD GENERALIZATION $\langle \mathsf{pop_d}, \mathbb{P} \rangle$, T3 AVERAGE ERROR $\langle \mathsf{pop_d}, \mathbb{E} \rangle$, T4 PER OPERATION $\langle \mathsf{op}, \forall \rangle$ (N), T5 SECURITY THREAT CONDITION $\langle \mathsf{op}, \forall \rangle$ (ADV), T6 LOCAL ROBUSTNESS $\langle \mathsf{in}, \forall \rangle$ (B). **D**: DISCHARGE. L_k: LEAST PATH WITH k OBLIGATIONS, LIFTS LISTED (THR: HAZARD COVERAGE ONLY). $\times$: CATEGORY ERROR. LAST COLUMN: WEIGHTED SHARE OF THE CLASS'S FULL-STUDY MEMBERS WHOSE LIFT ASSUMPTION IS EVIDENCED (SE OR SEC). GENERATED BY `TRANSFER_FULL.PY`.

Class	papers	type	T1 per FH	T2 ODD gen.	T3 average	T4 per op.	T5 security	T6 local rob.	ev. %
RS	130	$\langle \mathsf{in}, \forall_\alpha \rangle$	L_4 α,op,rate	L_3 α,op	L_3 α,op	L_4 α,dyn,hor	L_3 α,dyn,hor	L_2 α	26
BP	525	$\langle \mathsf{in}, \forall \rangle$	L_3 op,rate	L_2 op	L_2 op	L_3 dyn,hor	L_2 dyn,hor	L_1 thr	6
SMT	469	$\langle \mathsf{in}, \forall \rangle$	L_3 op,rate	L_2 op	L_2 op	L_3 dyn,hor	L_3 dyn,hor	L_1 thr	4
LIP	165	$\langle \mathsf{in}, \forall \rangle$	L_3 op,rate	L_2 op	L_2 op	L_3 dyn,hor	L_2 dyn,hor	L_1 thr	8
SEM	153	$\langle \mathsf{in}, \forall \rangle$	L_3 op,rate	L_2 op	L_2 op	L_3 dyn,hor	L_2 dyn,hor	L_1 thr	40
PIL	22	$\langle \mathsf{tr}, \forall \rangle$	L_3 hor,sup,rate	L_2 hor,sup	L_2 hor,sup	L_1 hor	L_2 hor	×	100
DTMC	50	$\langle \mathsf{pop_d}, \mathbb{E} \rangle$	×	×	**D**	×	×	×	0
CLR	212	$\langle \mathsf{tr}, \forall \rangle$	L_4 hor,sup,rate	L_3 hor,sup	L_3 hor,sup	L_2 hor	L_2 hor	×	10
CRL	261	$\langle \mathsf{tr}, \forall_\alpha \rangle$	L_5 α,hor,sup,rate	L_4 α,hor,sup	L_4 α,hor,sup	L_3 α,hor	L_2 α,hor	×	16
CRL^{f}	261	$\langle \mathsf{tr}, \forall \rangle$	L_3 hor,sup,rate	L_2 hor,sup	L_2 hor,sup	L_1 hor	L_2 hor	×	16
NCERT	221	$\langle \mathsf{tr}, \forall \rangle$	L_4 hor,sup,rate	L_3 hor,sup	L_3 hor,sup	L_2 hor	L_2 hor	×	10
CP	1183	$\langle \mathsf{op}, \mathbb{P} \rangle$	L_1 rate	**D**	**D**	×	×	×	35
CP^{f}	1183	$\langle \mathsf{pop_d}, \mathbb{P} \rangle$	L_1 rate	**D**	**D**	×	×	×	35
RTA	484	$\langle \mathsf{op}, \forall \rangle$	L_2 sup,rate	L_1 sup	L_1 sup	**D**	L_1 thr	×	31
MON	52	$\langle \mathsf{in}, \forall \rangle$	L_2 op,rate	L_1 op	L_1 op	L_2 dyn,hor	L_3 dyn,hor	L_1 thr	58
FALS	0	$\langle \mathsf{op}, \exists \rangle$	×	×	×	×	×	×	–
PACB	294	$\langle \mathsf{pop_d}, \mathbb{P} \rangle$	L_1 rate	**D**	**D**	×	×	×	31
PROB	45	$\langle \mathsf{in}, \mathbb{P} \rangle$	L_3 op,rate	L_2 op	L_2 op	×	×	×	37
OPREL	136	$\langle \mathsf{pop_d}, \mathbb{P} \rangle$	L_1 rate	**D**	**D**	×	×	×	16
GEN^{+}	891	$\langle \mathsf{pop_d}, \mathbb{P} \rangle$	L_1 rate	**D**	**D**	×	×	×	1
LUC^{+}	259	$\langle \mathsf{tr}, \mathbb{P} \rangle$	L_2 hor,rate	L_1 hor	L_1 hor	×	×	×	3
ALG^{+}	480	$\langle \mathsf{pop_d}, \mathbb{P} \rangle$	L_1 rate	**D**	**D**	×	×	×	4
GLOB^{+}	54	$\langle \mathsf{in}, \mathbb{E} \rangle$	×	×	×	×	×	×	62
IMPL^{+}	20	$\langle \mathsf{in}, \forall \rangle$	L_3 op,rate	L_2 op	L_2 op	L_3 dyn,hor	L_3 dyn,hor	L_1 thr	0
ARCH*	42	$\langle \mathsf{in}, \forall \rangle$	L_3 op,rate	L_2 op	L_2 op	L_3 dyn,hor	L_3 dyn,hor	L_1 thr	0
RLIM*	71	$\langle \mathsf{pop_d}, \mathbb{E} \rangle$	×	×	L_1 thr	×	×	×	0
DPG*	25	$\langle \mathsf{in}, \mathbb{E} \rangle$	×	×	×	×	×	×	0
APPROX*	19	$\langle \mathsf{in}, \forall_\alpha \rangle$	L_4 α,op,rate	L_3 α,op	L_3 α,op	L_4 α,dyn,hor	L_4 α,dyn,hor	L_2 α	33
GREG*	8	$\langle \mathsf{in}, \forall \rangle$	L_3 op,rate	L_2 op	L_2 op	L_3 dyn,hor	L_3 dyn,hor	L_1 thr	0
DSM*	8	$\langle \mathsf{in}, \forall \rangle$	L_3 op,rate	L_2 op	L_2 op	L_3 dyn,hor	L_2 dyn,hor	L_1 thr	100
XAI*	8	$\langle \mathsf{in}, \forall \rangle$	L_3 op,rate	L_2 op	L_2 op	L_3 dyn,hor	L_3 dyn,hor	L_1 thr	0
COMP*	13	$\langle \mathsf{in}, \forall \rangle$	L_3 op,rate	L_2 op	L_2 op	L_3 dyn,hor	L_2 dyn,hor	L_1 thr	0

directly; the evidence pack is a first instance. The missing step is $\mathsf{O}_{\mathsf{hor}}$ under realistic re-anchoring, including denial.

3) **Calibrated surrogate coverage** (PIL to T2, T4). Perception-in-the-loop results become transferable once $\mathsf{O}_{\mathsf{cov}}$ carries a confidence, as recent work does for contracts and semi-probabilistic controllers. The class is small (about 22 papers): an opening, not a backlog.
4) **Monitor coverage with confidence** (RTA and MON to T4). ASTM F3269 names coverage as a concern; 31 % of RTA guarantees evidence their monitor assumption, and only 2 % of lifted guarantees carry a confidence. Independence must be argued, not assumed.
5) **Hazard models that match the objective** (all T6 cells). Certified sets parametrized by the ODD, with an adversarial margin, would close the coverage obligation.

X. RELATED SOKS AND SURVEYS

Prior systematizations catalogue one level: certified robustness by technique, entirely at in [126], verification algo-

TABLE VII
THE RUNNING EXAMPLE AS A TYPED TRANSFER PATH, FROM THE GENERATED EVIDENCE PACK (RESULTS/EVIDENCE_PACK.JSON). STATUS: E EVIDENCED, P PARTIALLY EVIDENCED, S STATED, A ASSUMED, O OPEN.

Step	Lift	Type	Obligation and evidence	St.
$\theta \mapsto \Delta$	Lemma	$\langle \mathsf{in}, \forall \rangle$	proved; cap measured over 15 392 hops	E
$\Delta \mapsto \delta$	calibration	$\langle \mathsf{in}, \forall \rangle$	γ transfers to deployment: 3 real flights, one airframe, hover	P
$\delta \mapsto \rho$	$\mathsf{L}_{\mathrm{dyn}}$	$\langle \mathsf{tr}, \forall \rangle$	$\mathsf{O}_{\mathrm{lip}}$: local L measured on the checkpoint	E
$\rho \mapsto$ corridor	$\mathsf{L}_{\mathrm{hor}}$	$\langle \mathsf{op}, \forall \rangle$	$\mathsf{O}_{\mathrm{hor}}$: re-anchoring every T_c; delay adversary only	A
corridor $\mapsto$ MCR	$\mathsf{L}_{\mathrm{sup}}$	$\langle \mathsf{pop}_{\mathsf{d}}, \mathbb{P} \rangle$	$\mathsf{O}_{\mathrm{odd}}$: mission distribution named, simulated	S
MCR $\mapsto$ per FH	$\mathsf{L}_{\mathrm{rate}}$	$\langle \mathsf{pop}_{\mathsf{h}}, \mathbb{P} \rangle$	$\mathsf{O}_{\mathrm{dep}}$: not claimed	O
hazard	—	—	$\mathsf{O}_{\mathrm{thr}}$: delay adversary below threshold; denial, insider and monitor-input attacks not covered	S

rithms [127], safe learning-based control and learned certificates [82], [128], runtime assurance [99], and safety, trustworthiness, security and privacy of deep networks [129]–[131]. Tambon et al. review how to certify ML-based safety-critical systems and are our closest methodological precedent [132]. On the assurance side, the semantic gap between specification and learned behaviour [133], lifecycle assurance [134], safety-case patterns [135] and explicit residual doubt [136], [137] are established, as are the challenges of verified AI [138] and the case for combining statistical and formal guarantees [29]. None of these works types claims by what they quantify over, and none treats lifts as obligations whose deficit can be computed. That cross-level transfer analysis is our contribution. Our obligations map onto the assumption nodes and assurance claim points of goal structuring notation (GSN) [139], [140], so a transfer path can be exported as an argument fragment. Formally, the per-input level is a 2-safety hyperproperty [141], and the op/pop distinction is the distinction between path quantification and probabilistic quantification in probabilistic computation tree logic (PCTL) [142].

XI. LIMITATIONS AND THREATS TO VALIDITY

Every decision was made by an LLM. Screening, full-text coding and adjudication were done by LLM instances, a disclosed deviation from the registered human coding. Their agreement measures how unambiguous the codebook is, not how experts read the papers, and coders that share a training distribution may agree for the wrong reasons. A paper was screened out only when both screeners excluded it; the held-out half of the pilot set lost 2 of 27. The adjudicator shared a model with one coder and favoured its codes, so we state a conclusion only where each coder's estimate agrees. Every code is released, and a human re-coding regenerates the paper.

The corpus is a sample, and it is not saturated. Field-level numbers are weighted estimates from 587 coded papers in 6 strata. Intervals carry sampling error in coding and in screening, but not search error. The search retrieves 63 % of the pilot set. Snowballing stopped at its cap with every iteration above the 2 % stop threshold, so the expanded field is a citation neighbourhood, not the whole literature, and would keep growing; the type plane did not grow, which is the claim the stop rule protects. Seeds without references in OpenAlex (111 of 191 in the first iteration) contribute only citing papers. Full texts were limited to open-access copies, and 38 of 233 remainder records had none. If paywalled papers differ systematically, for example toward journal-only control results, the estimates are biased in an unknown direction. DBLP's venue indexes could not be swept.

The calculus is ours. The verdicts depend on the edge set. We chose edges that correspond to arguments in the corpus, and we release the code so that a reader can add, remove or re-weight an edge and regenerate the table. Unit weights treat all obligations alike. A deficit weighted by evidence kind is a natural refinement.

Types and sources. A class takes its modal type; members of other types are in Appendix D and the released codes. ED-324 and EASA Proposed Issue 03 were moving at our cut-off, and $22 - 14$ objectives rest on secondary sources, which we flag.

Construct validity. "Accounted for" (SE or SEC) is a strict reading of H2. Under a looser reading that counts any stated assumption, the share is 95 %. The gap H2 names is therefore one of evidence, not of disclosure.

XII. CONCLUSION

Certified robustness and airworthiness assurance do not fail to meet because one is too weak. They quantify over different things. Once guarantees and objectives carry types, the gap becomes computable, lifts bridge it, and each lift leaves an obligation behind. No chain of proofs reaches a probability objective without a statistical obligation, and runtime assurance moves the gap onto its monitor. The literature has moved toward population-level statistical guarantees, yet almost none take the rate lift to a per-hour figure, and few evidence the assumption regulators care most about: that the aircraft stays where the certificate holds. The evidence generator shows the alternative: every assumption listed as an obligation, and the boundary of the claim kept in the pack.

ETHICS CONSIDERATIONS

This work involves no human subjects and no personal data. It analyses published papers and public regulatory documents. Mapping guarantees to regulatory objectives could be misread as certification advice. We state throughout that it is not, and the evidence generator writes that disclaimer into every pack it emits. LLM screening used titles and abstracts, and LLM coding used open-access full texts of published papers. No unpublished material was processed, and the full texts are not redistributed.

Open Science

The protocol with every dated amendment, the search log, the candidate records, both screeners' decisions, the coding sample, both coders' codes, the adjudication log with rationales, the objective corpus, the calculus, the analysis scripts and the evidence generator are released in the anonymised artifact. Full texts are not, because they are third-party documents; their open-access sources are listed. Every number in this paper is a macro generated from those files.

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APPENDIX A
PROOFS

Proof of Proposition 1. $\succeq$ is the reflexive–transitive closure of the free edges, so it is reflexive and transitive. Each free edge either increases the scope index or moves the modality from $\mathbb{P}$ to $\mathbb{E}$, and no edge reverses either, so the closure has no cycles and is antisymmetric. Every edge into $\langle \mathsf{pop_h}, \mathbb{P}\rangle$ is $\mathsf{L_{rate}}$, which is not free, so no free path reaches it from $\langle \mathsf{in}, \forall\rangle$. No edge decreases scope, so no path at all leads back. The same argument applies to $\langle \mathsf{op}, \forall\rangle$ and $\langle \mathsf{pop_d}, \mathbb{P}\rangle$, through $\mathsf{L_{sup}}$. □

Proof of Theorem 1. Each lift is a valid rule: if its premise and its obligations hold, its conclusion holds. Let F_0 be the event that g fails and F_{O} the event that the evidence for O is invalid. Off $F_0 \cup \bigcup_{\mathsf{O}} F_{\mathsf{O}}$, induction along π shows that every intermediate conclusion holds. By the union bound, that event has probability at most δ_π. The error terms compose by substituting each rule's bound into the next. If the final type dominates τ_o, the objective follows by Definition 2. □

Proof of Proposition 2. The edges that change the modality from $\forall$ or $\forall_\alpha$ to $\mathbb{P}$ are $\mathsf{L_{op}}$, $\mathsf{L_{sup}}$ and $\mathsf{L_{mon}}$. They carry $\mathsf{O_{prof}}$, $\mathsf{O_{odd}}$ and $\mathsf{O_{mcov}}$, all statistical. L_α maps $\forall_\alpha$ to $\forall$, and every other edge preserves $\mathbb{P}$ or leaves it. Any path that ends at modality $\mathbb{P}$ from $\forall$ or $\forall_\alpha$ must therefore cross one of the three. The released code checks the claim exhaustively over all pairs of types. □

Proof of Proposition 3. (a) Let X_t indicate a hazardous decision. By linearity of expectation, $\mathbb{E}\sum_{t\le 3600f} X_t = 3600 f\, p$ for any joint law with marginals p. (b) Every hazardous run contains at least k errors, so the number of runs is at most the number of errors divided by k. Taking expectations gives the first bound. For the Markov chain, stationarity gives $\Pr(X_{t-1}=0, X_t=1) = \Pr(X_t=1, X_{t+1}=0) = p(1-\rho)$, and a run of length at least k starts at t with probability $p(1-\rho)\rho^{k-1}$. The function $(1-\rho)\rho^{k-1}$ is maximized at $\rho = (k-1)/k$, with value $(k-1)^{k-1}/k^k$. Under independence ($\rho = p$) the rate is $3600 f p^k (1-p)$. □

Proof of Proposition 4. (a) Under a Poisson process with rate λ, $\Pr(\text{no failure in } T) = e^{-\lambda T}$. The test rejects $\lambda \ge \lambda_0$ at level a if and only if $e^{-\lambda_0 T} \le a$. With $a = 0.05$, $T\lambda_0 = \ln 20 = 2.996$, so $T = 30, \ldots, 30\,000$ h give $\lambda_0 = 0.0999, \ldots, 9.99 \times 10^{-5}$. (b) With n independent Bernoulli frames, zero failures reject $p \ge p_0$ at level a when $(1-p_0)^n \le a$, that is, $n \ge \ln(1/a)/p_0$ to first order. With $p_0 = \lambda_0/(3600f)$ this is $n/(3600f) = \ln(1/a)/\lambda_0$ hours. □

Proof of Proposition 5. In the edge set, the only edges into $\langle \mathsf{op}, \forall\rangle$ are $\mathsf{L_{hor}}$ from $\langle \mathsf{tr}, \forall\rangle$ and the identity, so a direct discharge needs a guarantee already of that type. The Simplex theorem states safety on every execution in which the monitor flags each state outside the recoverable set within the switching latency, and the recovery controller is verified from every state it can be handed. These are $\mathsf{O_{mcov}}$, $\mathsf{O_{lat}}$ and $\mathsf{O_{rec}}$, and the monitor's input must not share the failure modes of the channel it guards, which is $\mathsf{O_{ind}}$. When the monitor's input is a learned perception output, the event "the monitor flags every hazardous state" has a probability over deployed inputs, which is of type $\langle \mathsf{pop_d}, \mathbb{P}\rangle$. □

Proof of Corollary 1. $\langle \mathsf{in}, \forall\rangle \xrightarrow{\mathsf{L_{op}}} \langle \mathsf{pop_d}, \mathbb{P}\rangle \xrightarrow{\mathsf{L_{rate}}} \langle \mathsf{pop_h}, \mathbb{P}\rangle$ carries $\{\mathsf{O_{prof}}, \mathsf{O_{dep}}\}$, and no path with fewer obligations exists, because every edge into $\langle \mathsf{pop_h}, \mathbb{P}\rangle$ is $\mathsf{L_{rate}}$ and no free edge leaves $\langle \mathsf{in}, \forall\rangle$. Any path through $\langle \mathsf{op}, \forall\rangle$ must enter it by $\mathsf{L_{hor}}$ from $\langle \mathsf{tr}, \forall\rangle$, which is reached from $\langle \mathsf{in}, \forall\rangle$ only by $\mathsf{L_{dyn}}$, and must leave it by $\mathsf{L_{sup}}$ and then $\mathsf{L_{rate}}$. That gives $\{\mathsf{O_{lip}}, \mathsf{O_{hor}}, \mathsf{O_{odd}}, \mathsf{O_{dep}}\}$. The released code reproduces both paths. □

APPENDIX B
SEARCH STRINGS

Block A (one query per phrase): "certified robustness", "certifiably robust", "formal verification", "reachability analy-

sis", "runtime assurance", "simplex architecture", "conformal prediction", "PAC-Bayes", "statistical model checking", "scenario approach", "safety verification", "provable guarantee", "perception contract", "safety certificate", "barrier certificate", "operational profile". Block B: ("neural network" OR "deep learning" OR "learning-enabled" OR "machine learning" OR "reinforcement learning"). Block C: ("aircraft" OR "UAV" OR "UAS" OR "drone" OR "quadrotor" OR "autonomous vehicle" OR "autonomous driving" OR "avionics" OR "airworthiness" OR "collision avoidance" OR "safety-critical"). Years 2017–2026. Revision 1 adds to block C: "aerospace", "autonomous system", "control system", "controller", "closed-loop", "cyber-physical", "hybrid system" and "robot"; and uses "formal analysis" only as a title filter in the venue sweeps, since as a search term it matches the CRediT contributor role printed in most journal articles. Database-specific syntax is generated by `search_corpus.py`, which logs every request.

Appendix C
Codebook Decision Rules

1) Randomized-smoothing confidence is $\forall_\alpha$, never $\mathbb{P}$.
2) Falsification and counterexample search are $\exists$. A zero-failure sampling campaign whose confidence enters the claim is $\mathbb{P}$.
3) If a certificate over a domain is checked by sampling and the sampling enters the claim, add L_{conc}.
4) A competition report inherits the scope of its benchmark category.
5) Runtime-assurance architectures with a conditional safety theorem are $\langle \mathsf{op}, \forall \rangle$ with L_{mon}.
6) Between two candidate scopes, choose the smaller and flag it.
7) (v1.1, CLR-1) Initial-set reachability with no disturbance input has perturbation "none".
8) (v1.1, PIL-1) A result over a generator's latent space is "semantic". A result over an estimated perception-error set is "distribution".
9) (v1.1, COMP-1) A learned trajectory or intent predictor is "estimation".

Appendix D
Guarantee Classes

Class	Members
RS	[1], [53]–[57]
BP	[3], [13]–[16], [48], [49]
SMT	[2], [12], [47], [66], [67]
LIP	[50]–[52]
SEM	[58]–[64], [94]
PIL	[10], [87]–[93]
DTMC	[95]
CLR	[4], [5], [17], [69]–[79]
CRL	[83]–[86]
NCERT	[80], [81]
CP	[113]–[117]
RTA	[27], [68], [96]–[98], [100]
MON	[101]–[103]
FALS	[105]–[107]
PACB	[109]–[111]
PROB	[119]–[121]
OPREL	[24], [25], [122]–[124]
GEN$^+$	learning-theoretic generalization and sample-complexity bounds outside PAC-Bayes (14 coded papers)
LUC$^+$	learned-uncertainty safe control: Gaussian-process bounds, chance constraints, adaptive barrier functions, learned reachable sets (6)
ALG$^+$	convergence and regret bounds of the learning algorithm (3)
GLOB$^+$	global robustness by Lipschitz-based optimization (2)
IMPL$^+$	verification of the executable implementation, e.g. timing (1)
ARCH*	guarantees by architectural construction: constraint layers, equivariance, structure-preserving models (5)
RLIM*	population-level robustness bounds and limits: distributionally robust risk, sensitivity, impossibility (7)
DPG*	differential privacy of training, and poisoning bounds derived from it (3)
APPROX*	approximation-error bounds against a reference: compression, approximate MPC, data-driven predictors (3)
GREG*, DSM*, XAI*, COMP*	global robustness by region enumeration, deterministic smoothing, sound explanations, compositional robustness (one each)

Appendix E
Acronyms

AC1	Gwet's first-order agreement coefficient
ACAS Xu	airborne collision avoidance system for unmanned aircraft
AINNCS	AI and neural-network control systems (ARCH-COMP category)
AMC	acceptable means of compliance
BP	bound propagation
CP	conformal prediction
DTMC	discrete-time Markov chain
EASA	European Union Aviation Safety Agency
EKF	extended Kalman filter
EUROCAE	European Organisation for Civil Aviation Equipment
FAA	Federal Aviation Administration
FH	flight hour
GSN	goal structuring notation
IBP	interval bound propagation
JARUS	Joint Authorities for Rulemaking of Unmanned Systems
LLM	large language model
MCR	mission-completion rate
ML / NN	machine learning / neural network
MLEAP	Machine Learning Application Approval (EASA project)
NNCS	neural-network control system
OD / ODD	operational domain / operational design domain
PABAK	prevalence-adjusted bias-adjusted kappa
PAC	probably approximately correct
PCTL	probabilistic computation tree logic
PRISMA	Preferred Reporting Items for Systematic reviews and Meta-Analyses
RS	randomized smoothing
RTA	run-time assurance
SAIL	specific assurance and integrity level
SMT	satisfiability modulo theories
SoK	systematization of knowledge
SORA	Specific Operations Risk Assessment
STL	signal temporal logic
TLS	target level of safety
UAS / UAV	unmanned aircraft system / unmanned aerial vehicle
VNN-COMP	International Verification of Neural Networks Competition